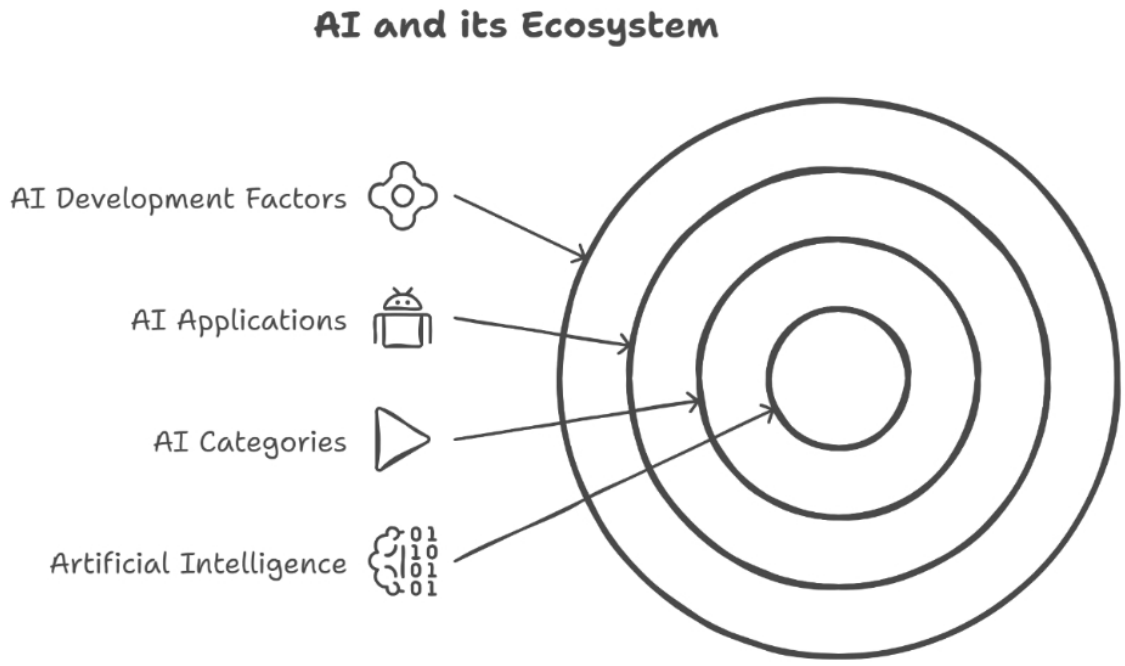


The overarching goal of AI is to develop systems that can operate autonomously, exhibit intelligent behavior, and refine their capabilities through the process of learning. Such advancements hold promise for revolutionizing fields such as healthcare, finance, education, and more. For example, in healthcare, AI-powered systems can analyze medical images with incredible accuracy, assist in diagnostics, and predict patient outcomes based on historical data.

AI is commonly classified into two broad categories:

- **Narrow AI**: This type of AI is designed to perform a specific task efficiently. Examples include voice-activated virtual assistants (e.g., Siri, Alexa), recommendation engines used by streaming services, and automated customer support systems. Narrow AI excels in predefined domains but lacks the ability to generalize across different tasks.
- **General AI**: This represents the pinnacle of AI research and development. General AI aims to replicate human intelligence in its entirety, allowing machines to perform any intellectual task that a human can accomplish. While still a theoretical